Weight decay sharpens, but does not move, the accessibility boundary

frozen-$d2$ setup, stage-2 sweep $\lambda \in [0, 10^{-1}]$ on block 2 + readout — 8 accessibility conditions × 9 λ × 3 seeds

Setup

Identical to the frozen accessibility experiment (see `accessibility_report.pdf`): 4 bits, full 2^4 cube, two ReLU residual blocks (N=1024, all init variances $1/N$), exact Walsh measurements. Stage 1 trains embedding + block 1 on the parents x_1x_2, x_3x_4 with $\lambda = 0$ to machine precision ($1{+}1 \to 2$ corner units, weighted singleton share 1.000), then is **frozen**, so weight decay can never touch the parent modes; ReLU homogeneity gives frozen representations with measured $R = Q_2/Q_1 \in \{0.12, \ldots, 0.66\}$ bracketing $R_c = 1/\sqrt{8} \approx 0.354$. Stage 2 trains a fresh block 2 + readout on $x_1x_2x_3x_4$, with the *same* weight decay λ on both (the minimum-norm-theory setting), $\lambda \in \{0, 10^{-5}, 10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$, identical stage-2 initialization across all (R, λ) within a seed. Circuits are classified from the Walsh decomposition of block-2 *pre-activations* (parent share vs. singleton share, contribution-weighted over the neurons carrying 95% of the $d4$ output). Every cell solves the task to $|A| \geq 0.70$; no (R, λ) in the range failed outright.

Findings

1. The boundary does not move. Across five decades of λ the transition stays pinned at R_c (Figs. 1, 2). Below R_c, decay pushes the compositional fraction *down* (at $R = 0.32$: $f_{2+2} = 0.052 \to 0.015$ as $\lambda: 0 \to 10^{-3}$); above R_c it pushes it *up* (at $R = 0.38$: $0.125 \to 0.517$; at $R = 0.44$: $0.221 \to 0.904$; at $R = 0.66$: $0.452 \to 0.994$). λ acts as a sharpener around a fixed pivot — exactly the static-theory prediction that $\eta_{\text{comp}}/\eta_{\text{direct}}$, hence the tie point, is λ-independent.

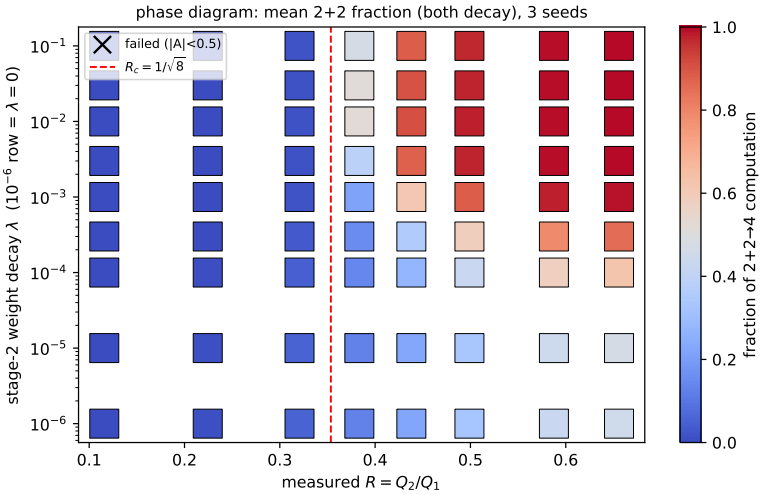


Figure 1: Phase diagram: mean 2+2 fraction on the (R, λ) grid (3 seeds; bottom row $\lambda = 0$). The boundary stays at $R_c = 1/\sqrt{8}$ (red) for all λ; color saturates with increasing λ on the right of the line and stays ≈ 0 on the left. No cell failed ($|A| < 0.5$ never occurred).

2. Sharpening saturates. The $f_{2+2}(R)$ curves steepen monotonically with λ and converge by $\lambda \approx 3{\cdot}10^{-3}$ (the $\lambda = 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}$ curves overlap): the crossover narrows from a gradual rise at $\lambda = 0$ (only 0.45 at $R = 0.66$) toward a near-step, $f_{2+2} \lesssim 0.02$ left of R_c vs. ≥ 0.88 for $R \geq 0.44$.

3. Why: decay forces the minimum-norm solution. At $R = 0.44$ the total stage-2 squared parameter norm collapses from 1034 ($\lambda = 0$, kitchen-sink interpolation) to 8.2 at $\lambda = 10^{-3}$ and 5.9 at 10^{-1}. At $\lambda = 0$ gradient descent keeps whatever mixture it first finds; under decay only the norm-efficient circuit survives, which is where the per-unit-norm comparison $\eta = |c_4|/|v|$ actually binds. Consistently, the mixed-neuron fraction at $R = 0.44$ drops from 0.10 ($\lambda = 0$) to 0.02 ($\lambda \geq 10^{-3}$): decay purifies the population into nearly binary 1^4 vs. 2+2 classes.

4. Suppression is ordered by accessibility, as $\lambda_c = e\eta$ predicts, but is never complete in range. Amplitude stays $|A| \approx 1$ through $\lambda = 3{\cdot}10^{-3}$ and erodes beyond 10^{-2} (Fig. 2, right), monotonically in R: at $\lambda = 10^{-1}$, $|A| = 0.70$ at $R = 0.12$ rising to 0.92 at $R = 0.66$ — the more accessible the parents, the more decay the circuit tolerates. Full suppression (λ_c) lies beyond 10^{-1} for every condition here, so the predicted "only the efficient circuit survives" window shows up in this range as amplitude ordering rather than outright failure of the direct route.

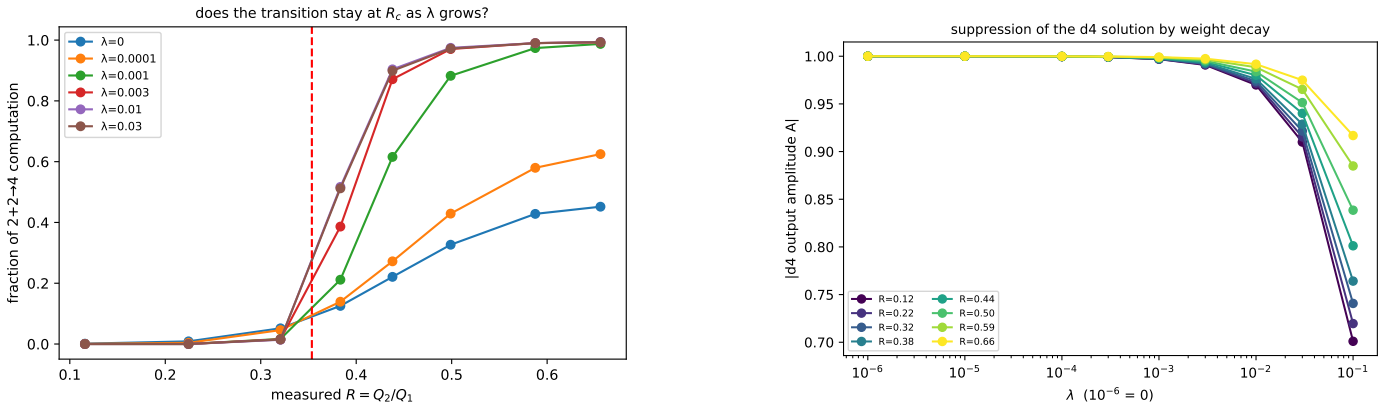


Figure 2: Left: f_{2+2} vs. measured R at fixed λ — all curves pivot at R_c; increasing λ steepens the transition without moving it, saturating by $\lambda \approx 3{\cdot}10^{-3}$. Right: $d4$ output amplitude vs. λ per condition — erosion starts near $\lambda \sim 10^{-2}$ and is ordered by R (higher accessibility → higher tolerated decay), the direction predicted by $\lambda_c = e\eta$.

Answers to the framing questions. (i) Accessibility still predicts circuit selection under weight decay — better than at $\lambda = 0$. (ii) Decay strengthens the preference for the higher-η circuit, saturating near $\lambda \sim 3{\cdot}10^{-3}$. (iii) The boundary remains at $1/\sqrt{8}$ throughout. (iv) Within $\lambda \leq 10^{-1}$ the inefficient circuit is weakened (amplitude ordering) but not yet extinguished; testing the full λ_c hierarchy needs larger λ or smaller parent amplitudes. The hidden-only decay control (`--hidden-only`) is implemented and not yet run.

Artifacts: `gdrive:parity/analysis_output/accessibility_wd/both/` (`results.csv`, `trajectories.csv` — per-1000-step fractions —, `block2_neurons.csv`, `stage1.json`, figures, this report); code: `analysis/accessibility/run_wd_experiment.py`. Companion: `accessibility_report.pdf`.